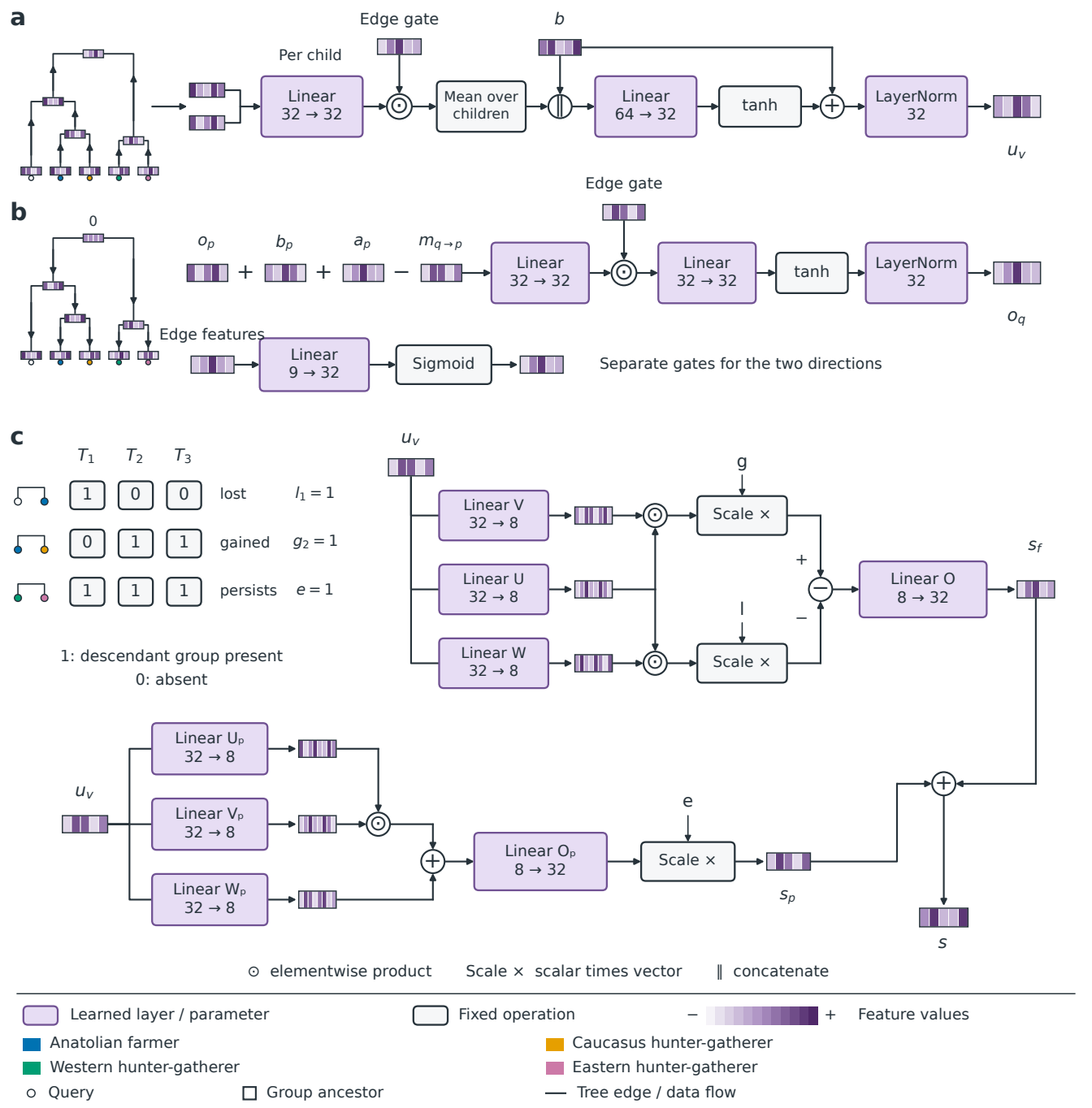
a
Edge gate
b
Per child
Linear 32 → 32
Mean over children
Linear 64 → 32
tanh
LayerNorm 32
u_v
b
0
o_p
b_p
a_p
$m_{q\to p}$
Edge gate
Linear 32 → 32
Linear 32 → 32
tanh
LayerNorm 32
o_q
Edge features
Linear 9 → 32
Sigmoid
Separate gates for the two directions
c
T_1 T_2 T_3
1 0 0 lost $l_1 = 1$
0 1 1 gained $g_2 = 1$
1 1 1 persists $e = 1$
1: descendant group present
0: absent
u_v
Linear V 32 → 8
Linear U 32 → 8
Linear W 32 → 8
g
Scale ×
l
Scale ×
+
−
Linear O 8 → 32
s_f
u_v
Linear U_p 32 → 8
Linear V_p 32 → 8
Linear W_p 32 → 8
Linear O_p 8 → 32
e
Scale ×
s_p
s
⊙ elementwise product
Scale × scalar times vector
‖ concatenate
Learned layer / parameter
Fixed operation
− +
Feature values
Anatolian farmer
Western hunter-gatherer
Caucasus hunter-gatherer
Eastern hunter-gatherer
Query
Group ancestor
Tree edge / data flow